Curriculum Vitae

Ayato Kitadai

Last updated: 2026/11/9

Education

- PhD Program, Department of Technology Management for Innovation, Graduate School of Engineering, The University of Tokyo, Apr. 2025 - Present
- Master's Program, Department of Technology Management for Innovation, Graduate School of Engineering, The University of Tokyo, Apr. 2023 - Mar. 2025
- Bachelor's Program, Department of Systems Innovation, Faculty of Engineering, The University of Tokyo, Apr. 2019 - Mar. 2023

Grants

- JSPS Research Fellowship for Young Scientists (DC1), Apr. 2025 - Present
- Publicly Offered Research Fund, Research Institute for Socionetwork Strategies (RISS), Kansai University, “The Effect of Generative AI Dialogue Intervention on Individual Decision-Making in a Dynamic CPR Game” (Co-Investigator), Jun. 2026 - Mar. 2028

Fellowships and Support

- WINGS GLAFS Excellent Research Assistant, Sep. 2023 - Mar. 2025

Teaching Experience

- Guest Lecturer, Interdisciplinary General Lecture (Introduction to AI), Meiji University, Fall 2025

Qualifications

- Teaching certificate, junior and senior high school mathematics

Research Areas

Service Engineering Mechanism Design

Game Theory Market Design

Experimental Economics

Behavioral Economics

Large Language Models

Multi-Agent Simulation

Peer-Reviewed Publications (15)

1. **A. Kitadai**, S.D.R. Lugo, Y. Tsurusaki, Y. Fukasawa and N. Nishino; “Can AI with High Reasoning Ability Replicate Human-like Decision Making in Economic Experiments?” *Group Decision and Negotiation*, 34(6), 1303-1326, 2025.
2. H. Sawazaki, X. Shang, S. Lee, **A. Kitadai** and N. Nishino; “Tenant allocation in shopping malls based on deferred acceptance algorithm,” *Procedia CIRP*, 138, 3-8, 2026.
3. S. Yoshimura, T. Kawasaki, **A. Kitadai** and N. Nishino; “Profitability of Slow Steaming Under Vessel Offshore Waiting Conditions,” *IEEE Big Data*, pp. 5947-5953, 2025.

4. T. Nakashima, T. Oyama, R. Wada, **A. Kitadai** and N. Nishino; “A Transdisciplinary Approach to Cross-Sectoral Collaboration for Climate Change Adaptation: Bridging Synergies and Trade-Offs,” *Advances in Transdisciplinary Engineering*, 76, 428-437, 2025.
5. **A. Kitadai**, S.D.R. Lugo and N. Nishino; “Matching Problems in Remanufactured Products for Product Service Systems in the Circular Economy,” *IFIP Advances in ICT*, 767, 246-257, 2025.
6. S. Lee, **A. Kitadai**, Ye Cheng and N. Nishino; “Transformation of Many-to-One to One-to-One Matching Problems Through Replication,” *IFIP Advances in ICT*, 767, 287-301, 2025.
7. **A. Kitadai**, K. Ogawa and N. Nishino; “Examining the Feasibility of Large Language Models as Survey Respondents,” *IEEE Big Data*, pp. 3858-3864, 2024.
8. Y. Takenoya, **A. Kitadai** and N. Nishino; “Accounting for Traffic Congestion: Heterogeneous Target Arrival Times, Uncertainty, and Monetary Incentives,” *IEEE Big Data*, pp. 3880-3888, 2024.
9. M. Kobayashi, **A. Kitadai**, U. Sato, M. Fujita and N. Nishino; “Real Traffic Analysis Around Large-Scale Shopping Mall Using Game Theory,” *IFIP Advances in ICT*, 730, 408-423, 2024.
10. M. Fujita, Z. Cheng, S.D.R. Lugo, **A. Kitadai**, Y. Dai, Y. Fukasawa and N. Nishino; “Analysis of Medical Data and Machine-Learning Algorithms from the Perspective of Public-Goods Models of Data-Provision Decision Making,” *Procedia CIRP*, 126, 3-8, 2024.
11. M. Fujita, Y. Dai, **A. Kitadai**, S.D.R. Lugo, Z. Cheng and N. Nishino; “Survey on Stakeholder Cooperative Behavior for Designing Voluntary Medical Data Provision Motivation Mechanisms,” *Procedia CIRP*, 126, 14-19, 2024.
12. **A. Kitadai**, S.D.R. Lugo, S. Lee, M. Fujita and N. Nishino; “Weighted Reward Allocation Mechanism for Data Collection Quality,” *Lecture Notes in Business Information Processing (LNBIP)*, 478, 161-174, 2023.
13. **A. Kitadai**, Y. Tsurusaki, Y. Fukasawa and N. Nishino; “Toward a Novel Methodology in Economic Experiments: Simulation of the Ultimatum Game with Large Language Models,” *IEEE Big Data*, pp. 3168-3175, 2023.
14. Z. Zhou, M. Kobayashi, U. Sato, K. Akashi, **A. Kitadai**, S. Sugihara, Y. Fukasawa, M. Fujita and N. Nishino; “Evaluating the alleviation of traffic congestion through Bus Rapid Transit using Multi-Agent Simulation,” *IEEE Big Data*, pp. 3237-3244, 2023.
15. M. Fujita, **A. Kitadai**, K. Sumikura and N. Nishino; “Toward the Utilization of Artificial Intelligence in Life Sciences: A Voluntary Medical Data Provisioning Model,” *IEEE Big Data*, pp. 3250-3257, 2022.

Working Papers (8)

1. Y. Hoshino, **A. Kitadai** and N. Nishino; “Do Matching Mechanisms Work with LLM Agents?,” arXiv:2606.03030, 2026.
2. **A. Kitadai**, S. Yoshimura, T. Nakashima, N. Torpo, R. Miratsu, N. Mizutani and N. Nishino; “‘Sail Fast, Then Wait’ in First-come, First-served Port Queues: Information Sharing for Sustainable Shipping,” arXiv:2602.01958, 2026. Under review.
3. **A. Kitadai**, T. Ito, Y. Nagoh, H. Takahashi, M. Fujita, S. Lee, F. Miyahara, T. Natsume and N. Nishino; “Decision Support System for Technology Opportunity Discovery: An Application of the Schwartz Theory of Basic Values,” arXiv:2602.11855, 2026. R&R at *Technology in Society*.

4. T. Nakashima and **A. Kitadai**; “Adaptation Beliefs and Defensive Over-Adaptation under Nonlinear Climate Risk.” Under review.
5. **A. Kitadai**, Y. Fukasawa and N. Nishino; “Do Rich Persona Data Improve the Calibration of LLM Agents to Human Behavior? Evidence from the Ultimatum Game.” Under review.
6. **A. Kitadai**, Y. Nakamoto and N. Nishino; “Do Generative AI Chatbots Increase Dominant-Strategy Play in Matching Mechanisms?” Under review.
7. **A. Kitadai** and Y. Nakamoto; “The Effect of AI-Mediated Institutional Communication on Decision-Making in a Dynamic Common Pool Resource Game.” Under review.
8. **A. Kitadai**, Y. Takenoya, S.D.R. Lugo and N. Nishino; “Coordinating Multi-Item Contributions through Weighted Proportional Reward Sharing under a Committed Budget.” Under review.

Conference Presentations (International, Oral, Peer-Reviewed) (9)

1. **A. Kitadai**, S. Yoshimura, T. Nakashima, N. Torpo, R. Miratsu, N. Mizutani and N. Nishino; “‘Sail Fast, Then Wait’ in First-come, First-served Port Queues: Information Sharing for Sustainable Shipping,” GDN 2026, Katowice, Poland, June 2026.
2. J. Zhang, **A. Kitadai** and N. Nishino; “The Dynamics of Coopetition: A Biform Game Analysis of Incentive Mechanisms in DevOps,” GDN 2026, Katowice, Poland, June 2026.
3. **A. Kitadai**, S. Yoshimura, T. Nakashima, N. Torpo, R. Miratsu, N. Mizutani and N. Nishino; “Information Asymmetry in FCFS Port Queues: A Game Theoretic Analysis of ‘Sail Fast, Then Wait’,” SMRC x MTEC/ICMASS 2026, Singapore, April 2026.
4. S. Yoshimura, T. Kawasaki, **A. Kitadai** and N. Nishino; “Analysis of Optimal Ship Navigation in Offshore Waiting Situations,” International Conference of Serviceology 2025, Stockholm, Dec. 3, 2025.
5. H. Cheng, S. Lee, H. Watanabe, **A. Kitadai**, Y. Nagoh, M. Fujita and N. Nishino; “How Can Barriers to Innovation Diffusion and Transition Be Identified from Bibliometric Data?,” R&D Management Conference 2025, July 1, 2025.
6. **A. Kitadai**, T. Nakashima, Y. Tsurusaki, R. Miratsu, N. Mizutani and N. Nishino; “Can rich information decrease anchorage waiting time? A game-theoretic analysis toward sustainable shipping,” 34th European Conference on Operational Research, Leeds, UK, June 23, 2025.
7. **A. Kitadai**, R. Ishikawa, S. Lee, J. Teng, K. Nishiyama and N. Nishino; “Consensus Formation in Physical, Virtual, and Interverse Spaces: Experimental Insights,” 25th International Conference on Group Decision and Negotiation, Zaragoza, Spain, June 16, 2025.
8. H. Sawazaki, X. Shang, S. Lee, **A. Kitadai** and N. Nishino; “Tenant allocation in shopping malls based on deferred acceptance algorithm,” 18th CIRP International Conference on Intelligent Computation in Manufacturing Engineering, Naples, Italy, July 10, 2024.
9. **A. Kitadai**, S.D.R. Lugo, Y. Tsurusaki, Y. Fukasawa and N. Nishino; “Can AI with High Reasoning Ability Replicate Human-like Decision Making in Economic Experiments?,” 24th International Conference on Group Decision and Negotiation & 10th International Conference on Decision Support System Technology, Faculty of Engineering of the University of Porto, Porto, Portugal, June 3, 2024.

Conference Presentations (Domestic, Oral) (27)

1. N. Nishino, S. Lee and **A. Kitadai**; “Mechanism Design of Communication Environments in the Interverse: A Proposed Framework,” JIMA Fall Conference 2026, Arcrea Himeji, Himeji,

October 25, 2026.

2. **A. Kitadai** and N. Nishino; “Analysis of Anchorage Waiting Using a First-Come, First-Served Queueing Game and the Potential for Mitigation through Information Sharing,” JIMA Fall Conference 2026, Arcrea Himeji, Himeji, October 24, 2026.
3. **A. Kitadai**, Y. Kumagai and R. Yasunaga; “Bounded Rationality in Large Language Models: Price-Quality Heuristics and Self-Control,” JSAI 2026: The 40th Annual Conference of the Japanese Society for Artificial Intelligence, Gunma and online, June 2026.
4. Y. Hoshino, **A. Kitadai** and N. Nishino; “Service Design for Proxy Negotiation Markets with AI Agents: Validating the Effectiveness of Matching Mechanisms,” JIMA Spring Conference 2026, Hakodate, Mar. 4, 2026.
5. Y. Takenoya, **A. Kitadai** and N. Nishino; “A Screening Model of Upfront Contracts for Agents Who Ignore Convex Inequalities,” JIMA Spring Conference 2026, Hakodate, June 2026.
6. **A. Kitadai** and N. Nishino; “The Effect of Introducing a Free-Dialogue Chatbot for Supporting Understanding of Matching Mechanisms on People's Decision-Making,” JIMA Spring Conference 2026, Hakodate, June 2026.
7. **A. Kitadai**, Y. Nakamoto and N. Nishino; “Do Generative AI Chatbots Increase Dominant-Strategy Play in Matching Mechanisms?,” 29th Experimental Social Science Conference, Kochi, May 2026.
8. Y. Hoshino, **A. Kitadai** and N. Nishino; “Service Design for Proxy Negotiation Markets with AI Agents: Validating the Effectiveness of Matching Algorithms,” 14th Annual Conference of the Service Science Society of Japan, Mar. 2026.
9. Y. Yoshida, **A. Kitadai** and N. Nishino; “Game-Theoretic Analysis of Dynamic Production Planning and Core Collection Strategies in Commercial Vehicle Remanufacturing,” 14th Annual Conference of the Service Science Society of Japan, Mar. 4, 2026.
10. S. Yoshimura, T. Kawasaki, **A. Kitadai** and N. Nishino; “Profitability of Slow Steaming Under Vessel Offshore Waiting Conditions,” 14th Annual Conference of the Service Science Society of Japan, Mar. 5, 2026.
11. Y. Yoshida, **A. Kitadai** and N. Nishino; “Environmental Impact of Remanufacturing in the Commercial Truck Industry,” JIMA Fall Conference 2025, Oct. 19, 2025.
12. **A. Kitadai**, R. Ishikawa, S. Lee, J. Teng, K. Nishiyama and N. Nishino; “New Perspectives on Service Design for the Virtual Economy,” 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 5, 2025.
13. H. Watanabe, **A. Kitadai**, Y. Nagoh, S. Lee, T. Natsume, F. Miyahara, M. Fujita and N. Nishino; “A Decision Support Tool for New Product Development Using LLMs and Matching Models,” 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 5, 2025.
14. Z. Zhou, **A. Kitadai**, U. Sato, Y. Takenoya, M. Fujita and N. Nishino; “Evaluation of Bus Rapid Transit Introduction for Traffic Congestion Mitigation,” 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 5, 2025.
15. Y. Takenoya, **A. Kitadai** and N. Nishino; “Analysis of Monetary Incentives for Congestion Mitigation in a Departure-Time Choice Model,” 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 6, 2025.
16. T. Nakamura, **A. Kitadai** and N. Nishino; “Theoretical Analysis and Economic Experiment on Cooperative Behavior in Data Collection: A Nonlinear Public Goods Game Under Incomplete

Information," 13th Annual Conference of the Service Science Society of Japan, Osaka, Mar. 6, 2025.

17. **A. Kitadai**, Y. Fukasawa and N. Nishino; "Simulation of the Ultimatum Game with Large Language Models: Reproducibility of Experimental Results Through Bias Adjustment," 18th Annual Meeting of the Behavioral Economics Society, Osaka, Dec. 7, 2024.
18. **A. Kitadai** and N. Nishino; "Substitutability of Large Language Models as Survey Respondents," JIMA Fall Conference, Shimane, Oct. 26, 2024.
19. Y. Takenoya, **A. Kitadai** and N. Nishino; "Numerical Analysis of Traffic Evaluation Incorporating Heterogeneous Target Arrival Times," JIMA Fall Conference, Shimane, Oct. 26, 2024.
20. Z. Zhou, **A. Kitadai**, U. Sato, Y. Takenoya and N. Nishino; "Evaluation of Effective Bus Rapid Transit Introduction for Traffic Congestion Mitigation," JIMA Fall Conference, Shimane, Oct. 26, 2024.
21. M. Kobayashi, **A. Kitadai**, Z. Zhou, M. Fujita and N. Nishino; "Traffic Analysis for Congestion Mitigation Around Large-Scale Shopping Facilities," 12th Annual Conference of the Service Science Society of Japan, Tokyo, Mar. 6, 2024.
22. Z. Cheng, **A. Kitadai**, S.D.R. Lugo, N. Nishino, M. Fujita and L. Zhang; "Medical Data Classification and Provision Willingness Analysis Based on Public Goods Perspective," 12th Annual Conference of the Service Science Society of Japan, Tokyo, Mar. 7, 2024.
23. Y. Cheng, H. Sawazaki, N. Nishino, **A. Kitadai** and S. Lee; "Transformation of Many-to-One Matching Problems into One-to-One Matching Problems Through Replication," 12th Annual Conference of the Service Science Society of Japan, Tokyo, Mar. 7, 2024.
24. **A. Kitadai**, Y. Tsurusaki, Y. Fukasawa and N. Nishino; "Simulation of the Ultimatum Game with Large Language Models: Toward a New Methodology in Economic Experiments," 17th Annual Meeting of the Behavioral Economics Society, Kochi, Dec. 9, 2023.
25. **A. Kitadai**, M. Fujita and N. Nishino; "Designing a Mechanism for Smooth Data Collection," 11th Annual Conference of the Service Science Society of Japan, Kyoto, Mar. 9, 2023.
26. T. Ito, N. Nishino, H. Takahashi, **A. Kitadai**, M. Fujita, Y. Nagoh, T. Natsume and F. Miyahara; "How Do Technologies Create Value? A Function-Value Matching Model and Case Study," 11th Annual Conference of the Service Science Society of Japan, Kyoto, Mar. 9, 2023.
27. M. Fujita, **A. Kitadai**, K. Sumikura and N. Nishino; "Designing Mechanisms to Facilitate Medical Data Provision for AI Utilization in Life Sciences," 21st SIG-BI, Japanese Society for Artificial Intelligence, Kanagawa, Sep. 18, 2022.

Awards (8)

1. Student Encouragement Award, 14th Annual Conference of the Service Science Society of Japan (2026): Y. Hoshino, **A. Kitadai** and N. Nishino, "Service Design for Proxy Negotiation Markets with AI Agents: Validating the Effectiveness of Matching Algorithms."
2. 2026 GDN Springer Best Student Paper Award (2026)
3. Head of Department Award (Academic Achievement), Dept. of Technology Management for Innovation, Graduate School of Engineering, The University of Tokyo (2025)
4. 2024 GDN Springer Best Student Paper Award (2024)
5. Outstanding Conference Award and Arai Award, 12th Annual Conference of the Service Science Society of Japan (2024)

6. Poster Report Encouragement Award (General Division), Behavioral Economics Society (2023)
7. 2023 GDN Springer Best Student Paper Award (2023)
8. Outstanding Conference Award, 11th Annual Conference of the Service Science Society of Japan (2023)

Talks and Other Activities (15)

1. **Ayato Kitadai**, “Matching Mechanisms and AI,” FY2026 Second Online Research Seminar, Japan Industrial Management Association (JIMA), “Developments in Service Mechanism Design Research” (Service Division), online (Zoom), September 25, 2026 (invited).
2. **Ayato Kitadai**, “Toward the Use of Simulation in Experimental Economics: AI Agents and Human Decision-Making,” RISS Data Creation Unit Workshop, Research Institute for Socionetwork Strategies (RISS), Kansai University, Multimedia Laboratory (6F), Senriyama Campus, Suita, Osaka, September 1, 2026 (invited).
3. **A. Kitadai**, Y. Nakamoto and N. Nishino; “Do Generative AI Chatbots Increase Dominant-Strategy Play in Matching Mechanisms?,” International Workshop on Human-AI Interaction, ISER, Osaka, March 24, 2026 (invited).
4. Y. Hoshino, **A. Kitadai** and N. Nishino; “Do Matching Mechanisms Work with LLM Agents?,” International Workshop on Human-AI Interaction, ISER, Osaka, March 23, 2026.
5. **Ayato Kitadai**, “AI as Part of Mechanism Design: Evidence from Laboratory Experiments,” Behavioral and Experimental Economics Seminar (BEES), Waseda University, June 16, 2026.
6. **Ayato Kitadai**, “Exploring Various Methods for Generative Agents to Replicate Results in Economic Experiments of the Ultimatum Game,” HSE Comsoc seminar, DeCAn, February 12, 2026 (invited).
7. **A. Kitadai**, Y. Nakamoto and N. Nishino; “Do Generative AI Chatbots Increase Dominant-Strategy Play in Matching Mechanisms?,” Kansai University RISS Workshop, February 28, 2026.
8. Y. Hoshino, **A. Kitadai** and N. Nishino; “Do Matching Mechanisms Work with LLM Agents?,” Kansai University RISS Workshop, February 28, 2026.
9. **Ayato Kitadai**, “Substitutability of Human Behavior by LLM Agents: When and When Not to Use Silicon Sampling,” invited seminar, Analytical Marketing Research Division, Japan Marketing Science Association, October 7, 2025.
10. **Ayato Kitadai**, “AI-driven ‘Human-like’ Decision Making: Utilizing Insights and Data from Behavioral Economics,” Japan Association for Evolutionary Economics Online Lecture Series, July 19, 2025 (invited).
11. **Ayato Kitadai**, “Comment on ‘Keizai Seminar’ August-September 2024 Issue,” Keizai Seminar December 2024 - January 2025 Issue.
12. **Ayato Kitadai**, “Economic Experiments with AI as Subjects,” Joint Research Project Management Support Workshop (Kyoto Sangyo University), August 1, 2024 (invited).
13. **Ayato Kitadai**, “Exploring Various Methods for Generative Agents to Replicate Results in Economic Experiments of the Ultimatum Game,” 1st CEE-M Workshop on Artificial Intelligence in Research, Center for Environmental Economics - Montpellier (CEE-M), Faculty of Economics, Richter campus, Montpellier, France (Amphi 319), hybrid; online presentation, July 9, 2024 (invited).

14. **Ayato Kitadai**, “Decision-making Analysis of Data Provision as a Public Goods Model,” 2nd Mechanism Synthesis Research Seminar, March 1, 2024 (invited).
15. **Ayato Kitadai**, “Toward a Novel Methodology in Economic Experiments,” Waseda Game Theory and Experimental Economics Seminar, January 26, 2024 (invited).